Track: Quantitative Systems & Algorithmic Treasury

Challenge: The Reserve Juggler

1. The Premise

A modern digital bank or investment platform processes $20,000,000 in customer deposits, withdrawals, and merchant settlements every single hour.

Managing this volume of capital involves a critical institutional dilemma:

- **The Cash Buffer:** Money left idle in operational checking reserves earns 0% interest, but guarantees that whenever a user withdraws funds or completes a transaction, the cash is immediately on hand.
- **The Yield Vaults:** Deploying money into financial markets generates substantial interest, but every market asset introduces frictions: lockup windows, early-exit fees, duration risk, or equity volatility.

Holding too much cash in the buffer bleeds thousands of dollars every day in lost opportunity cost. Conversely, leaving the buffer thin to chase maximum yield risks catastrophic early-liquidation penalties or emergency central-bank overdraft loans when a sudden wave of customer withdrawals arrives (e.g., payday cycles, holiday shopping, or unexpected liquidity runs).

2. The Asset Landscape

Your platform has access to four distinct liquidity and investment tiers:

Tier	Instrument	Return Profile	Liquidity Friction & Risk
Tier 0: Cash Buffer	Operational Checking Reserves	0.0% APY	**100% Liquid.** Zero risk, zero transaction cost. Instantly absorbs customer payouts.
Tier 1: Overnight Repo	Central Bank Repurchase Facility	Floating APY (~4.3%)	**Time-Locked.** Swept into at 17:00, matures at 08:00 next morning with overnight interest. Cracking this vault early to cover midnight withdrawal surges incurs a 0.5% early-break fee.
Tier 2: 2Y Sovereign Bonds	Short-Term Government Debt	Floating Yield (~4.8%) + Capital PnL	**Duration Risk.** Bond prices fluctuate inversely with market yields ($dP \approx -D_{mod} \cdot dy$). Liquidating bonds mid-day incurs a 2.0% market haircut.
Tier 3: Growth Equities	Blue-Chip Equity Index	~10.0% Expected APY (Dynamic market index)	**Market Volatility.** Subject to continuous price fluctuations ($\sigma = 18\%$) and drawdowns. Emergency fire-sale liquidations carry a severe 15.0% slippage fee.

3. Provided Data: `treasury_4tier_dataset.csv`

Participants are provided 8,760 hours (one simulated calendar year) of synchronous banking and financial market data:

- `timestamp`: Hourly datetime index (YYYY-MM-DD HH:MM:SS).
- `deposits_usd`: Inbound customer deposits credited during the hour.
- `withdrawals_usd`: Customer payout and debit obligations clearing during the hour.
- `repo_sweep_window_open`: Flag (True/False) indicating the daily 17:00 repo entry cutoff.
- `repo_sweep_mature`: Flag (True/False) indicating the daily 08:00 repo release and settlement.
- `tier1_repo_rate_apy`: Current annualized policy rate for the repo facility.
- `tier2_bond_yield_pct`: Current yield-to-maturity of 2Y sovereign debt.
- `tier2_bond_price_index`: Clean mark-to-market bond index reflecting duration shifts.
- `tier3_stock_index_price`: Current market price of the equity index.
- `tier3_stock_volatility_pct`: Rolling annualized equity volatility index.